

# AI for Manufacturing Engineering

AI techniques for smart and efficient manufacturing systems.

|                 |                                  |          |                   |
|-----------------|----------------------------------|----------|-------------------|
| Programme level | Graduate, Postgraduate           | Duration | 14 weeks          |
| Credits         | 4                                | Delivery | Classroom, Hybrid |
| Subject area    | AI for Manufacturing Engineering | Updated  | 2026-07-21        |

## Course overview

An applied course linking artificial intelligence with manufacturing engineering decisions, process performance, quality, maintenance and production systems.

**Intended audience:** Manufacturing, industrial, mechanical and engineering management students and professionals.

**Prerequisites:** Basic manufacturing systems knowledge and introductory statistics.

## Learning outcomes

- Identify manufacturing decisions that can benefit from AI.
- Prepare sensor and production data for analysis.
- Evaluate quality, maintenance and process applications.
- Connect AI models with manufacturing workflows and controls.
- Design an implementation proposal for a smart manufacturing use case.

## Module structure

|           |                                                                                                                    |
|-----------|--------------------------------------------------------------------------------------------------------------------|
| <b>01</b> | <b>AI opportunities in manufacturing</b><br>Production, quality, maintenance, energy and flow.                     |
| <b>02</b> | <b>Industrial data foundations</b><br>Sensors, historians, traceability and data quality.                          |
| <b>03</b> | <b>Predictive quality and maintenance</b><br>Failure patterns, anomaly detection and decision thresholds.          |
| <b>04</b> | <b>Process optimisation and intelligent control</b><br>Operational recommendations and closed-loop considerations. |
| <b>05</b> | <b>Smart manufacturing implementation</b><br>Architecture, workforce, risk and value realisation.                  |

## Assessment pattern

| Assessment component        | Weight |
|-----------------------------|--------|
| Manufacturing data exercise | 25%    |
| Applied AI case analysis    | 30%    |
| Smart manufacturing project | 45%    |

## Delivery note

The module sequence, class activities and assessment details may be adapted for programme schedules, cohort experience and institutional requirements. The course remains focused on practical application, structured judgement and professional communication.